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Optimization model for designing personalized tourism packages

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ABSTRACT

The tourism supply chain aims at satisfying the needs of the tourists based on their preferences. However, the preference of each tourist may be different. Some tourists prefer to optimize a single criterion, while others prefer to optimize conflicting multiple-criteria. The tourism service provider can hardly offer the tourists with the itinerary according to their precise preferences. This paper proposes a multi-objective optimization framework based on which tourists can generate itineraries according to their preferences. A mathematical model is presented, which is multi-objective and NP-hard. Consequently, four *meta*-heuristic algorithms, namely none-dominated sorting genetic algorithm versions II (NSGA-II) and III (NSGA-III), multi objective grey wolf optimization, and multi objective imperialist competitive algorithm are developed. The proposed method helps the tourists to compare different combinations of activities and select the one that best suits their preferences. The model is tested on a small-scale real case pertaining to the Sultanat of Oman. Thereafter, the performances of the proposed algorithms were evaluated on large scale problems. The result shows that NSGAs outperformed other algorithms. NSGA-II outperformed its NSGA-III counterpart in small instances. Surprisingly, as the size of the problem increases, the efficiency of NSGA-II decreases while that of NSGA-III increases. In large instances, NSGA-III outperformed NSGA-II.

1. Introduction

The Organization for Economic Co-operation and Development (OECD) defines tourism as the activities or events because of which people travel and stay in for some time outside their original environment for official purposes or leisure for no more than one continuous year (WTO, 2001). The tourism industry involves a combination of people, providers, and organizations that integrate their efforts with the aim of serving tourists. It is one of the fastest growing industries and is an increasingly important source of income, employment, and wealth in many countries (Neto, 2003). The industry continuously develops in terms of infrastructure and services provided to increase the satisfaction of the tourists. According to the World Travel and Tourism Council (WTTC), the travel and tourism sector generates about 10 % of global GDP, which is $7.6 trillion and it is projected to reach around 11 % of the global GDP by 2026 (WTTC, 2015). This indicates that the tourism industry is a major contributor to the GDP globally and is a potential industry for developed and developing countries. Moreover, the industry contributes significantly to the employment and economic development of the touristic destinations and the surrounding areas (Rebollo & Baidal, 2003).

The tourism industry involves a set of parties to organize the operations of the tourism supply chain (TSC) and satisfy the requirements of tourists (Zhang et al., 2009, Mancini et al., 2022). Intense competition forces various parties to cluster and cooperate in supply chains to enhance their agility and improve their performance (Sigala, 2008; Piya et al., 2020). The parties in TSC are the service providers that serve the industry by providing their services and products to the end-users. The service providers are clustered into various groups, which include transportation, accommodation, entertainment, heritage and culture, and food. The end users are the tourists that are either domestic or international travelers Due to the nature of the tourism supply chain, the tourism product cannot be stored for future use since the demand is highly uncertain. The uncertainty of the demand is mainly due to the

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different tourists' preferences and interests, and it is seasonally affected. Such uncertainty creates challenges for the service providers in terms of designing innovative tourism products. Significant cooperation between multiple service providers in a TSC is detrimental to the success of a product and contributes to the optimization of the whole supply chain (Szpilko, 2017). The preparation of a tourism product often involves many parties providing various tourism related products and services. Due to the frequent interaction between the different parties, the structure of TCS is highly complex.

The main goal of the tourism industry is to satisfy the needs of the end-users based on their different preferences with respect to transportation, activities, and places of interest at the lowest costs. This creates conflicts between different groups of tourists who may seek different interests and activities at various locations. One of the important functions of the TSC is to interconnect different service providers to offer the best service to the end-users. Tour operators or travel agencies play a significant role in offering different service packages and tour itineraries to the tourists in coordination with the other service providers. An itinerary represents a planned set of activities and travel route the tourist is going to follow during his/her journey (Wong & McKercher, 2012). The aim of recommending an itinerary is to provide a sequence of visiting different locations while performing various activities in these locations within a given time frame and cost (Yochum et al., 2020). Most of the travel agencies generate general-purpose itineraries that may not completely take into account the preferences of tourists. This leads to causing wastes (or unsatisfaction) with respect to time spent in travelling, distance voyaged, trip costs, and number of activities, thereby lowering the satisfaction of the tourists. According to Huang et al. (2021), to remain competitive, the tourism service providers should be able to identify tourist preferences and customize their travel itineraries.

This research aims at designing personalized tourism packages by allowing the tourists the ability to select the preferred set of activities and defining the major inputs in the generation of the tourism package. The developed model helps in solving the problems related to the tourism planning that the tourists usually face while preparing for their trips and in avoiding waste in terms of time, distance, and cost. The remaining sections of the paper is organized as follows: Section 2 summarizes the relevant literature references in the field and highlights the research gaps. Section 3 discusses the conceptual framework of this research work. Section 4 presents the mathematical model that we developed to solve the aforementioned personalized tourism problem. Section 5 is devoted to four *meta*-heuristic algorithms, namely nonedominated sorting genetic algorithm versions II (NSGA-II) and III (NSGA-III), multi objective grey wolf optimization (MOGWO), and multi objective imperialist competitive algorithm (MOICA), which is proposed to solve large scale problems under study. Section 6 discusses the numerical analysis and the main findings of the study. Finally, the research concludes in Section 7 along with possible future research directions.

2. Literature review

The Optimum Personalized Tourism Package (OPTP) has been usually modelled in terms of the well-known Orienteering Problem (Golden et al., 1987). The Orienteering Problem is originated from the homonymous competition, in which the participants visit a set of checkpoints during a given amount of time. Each checkpoint is associated with a known score value and the participants aim to visit a subset of the checkpoints that will guarantee them the accumulation of maximum score (Ke et al., 2016). The winner is the participant who succeeds in achieving the highest score through not only selecting the appropriate subset of checkpoints but also identifying the most efficient sequence of their tour (Yu et al., 2019). The Orienteering Problem has been used to represent a wide set of applications, and researchers have developed many variants over the years in order to face the challenges arising in different contexts (Campbell et al., 2011). Gunawan et al. (2016) reviews the most recent variants of the problem, enumerates the solution methods for their solution and describes the most relevant related applications.

The Orienteering Problem has often been the optimization engine for defining personalized tourism packages. Table 1 summarizes the contributions of the related papers that have been published on the topic during recent years. The table highlights how our study follows most of the papers in being application-oriented and confirms that all the studies incorporate time-restrictions within their developed models. However, it is clear that only a few studies explicitly considered the budget cap on the packages to be defined. More importantly, Table 1 shows that only a few papers have recognized the multi-criteria nature of the OPTP and have proposed a multi-objective optimization model for its solution.

More specifically, Yu et al. (2015) solved the OPTP that employs a robot to collect useful information regarding a set of touristic points of interest (POIs). Their work considered two different objectives to be achieved by the robot: first, called budget-minimizing tourist, which seeks to minimize the time spent at the selected POIs and second, called reward-maximizing tourist, which attempts to maximize the total information gained from the POI visits. It is to be noted that even though the authors considered two objectives to be optimized, they used them separately within their optimization framework without ever defining a multi-objective model, as in our case. The study also developed a solution approach called anytime planning algorithm and applied it to generate a day tour among 20 potential POIs spatially distributed in Istanbul.

De Falco et al. (2016) developed an evolutionary algorithm to plan personalized touristic packages but focused only on walking tours within a confined urban zone. The authors defined and optimized five objective functions including maximizing the score of the defined tours and the number of POIs to be visited and minimizing the distance covered as well as any temporal deviation with respect to the planned schedule. They applied their approach to define walking tours in the city of Naples and carried out extensive experiments to assess the effect of the most relevant problem's parameters. However, their study did not take into account the budget restriction and an optimization model, but defined only the objectives used as fitness criteria for their population-based method. Earlier, other works such as Vansteenwegen et al. (2011), Cotfas et al. (2011), and Gavalas et al. (2015) also dealt with the definition of personalized walking tours but involved less features with respect to De Falco et al. (2016) and, consequently, with respect to our study.

Lim et al. (2018) proposed a tour recommendation framework that incorporates a bi-objective optimization model. The first objective maximizes the tourists' preferences with respect to the POIs to be visited, whereas the second one maximizes their popularity score. The paper developed an Orienteering Problem-based solution method and applied the suggested approach to define recommended tours in 10 different big cities sparsed around the globe. The last paper implementing a multi-objective approach that we are aware of is due to Yochum et al. (2020). However, even this work does not develop a multi-objective model in the commonly known sense, but incorporates a multi-criteria fitness function within a genetic algorithm method. The study considered the following criteria: the total number of POIs, the number of mandatory POIs to be included in the tour, the POIs' popularity, their entrance cost, and their overall rating. The authors proposed an adaptive genetic algorithm to solve the OPTP problem and used it to identify touristic itineraries for the following six cities: Budapest, Edinburgh, Toronto, Glasgow, Perth, and Osaka.

The above literature review has identified several research gaps that we try to address in this work. Specifically, our study:

- Proposes explicitly the first ever multi-objective optimization model for the solution of the OPTP problem with operational constraints;
- Incorporates explicitly the budget restriction on the touristic packages to be identified;

Table 1
Recent papers related to the use of the Orienteering Problem to solve the OPTP Problem.

Reference	Application	Flexible Start/End Nodes	Multi-Objective Model	Time Restricted	Budget Restricted	Solution Approach
Sylejmani et al. (2014)	Prishtina			✓		Meta-heuristic
Herzog and Wörndl (2014)				✓		Heuristic
Verbeeck et al. (2014)		✓		✓		Meta-heuristic
Malucelli et al. (2015)	Trebon region			✓	✓	Exact
Yu et al. (2015)	Istanbul	✓	✓ (bi-)	✓		Exact & heuristic
Gavalas et al. (2015)	Athens and Berlin			✓		Heuristic
De Falco et al. (2016)	Naples	✓	✓	✓		Meta-heuristic
Brito et al. (2017)				✓		Meta-heuristic
Lim et al. (2018)	Several cities		✓ (bi-)	✓	✓	Heuristic
Mancini and Stecca (2018)	Mediterranean sea			✓		Meta-heuristic
Hapsari et al. (2019)	Besuki			✓		Exact
Expósito et al. (2019)				✓		Meta-heuristic
Yochum et al. (2020)	Several cities		✓	✓		Meta-heuristic
Tenemaza et al. (2020)	Paris, Rome, New York	✓		✓		Meta-heuristic
Persia et al. (2020)	Bolzano			✓		Heuristic
Urrutia-Zambrana et al. (2021)	Three Spanish cities			✓		Meta-heuristic
Ruiz-Meza and Montoya-Torres (2021)			✓ (bi-)	✓ (bi-)		Exact
This Study	Muscat	✓	✓	✓	✓	Exact and *meta*-heuristic

- Applies the developed approach for the first time to build OPTP in the city of Muscat as an emerging touristic destination that still needs to be discovered by the international tourists;
- Develops four novel and well-known *meta*-heuristic algorithms—NSGA-II, NSGA-III, MOGWO, and MOICA—to solve the problem and compare their performances to identify the best *meta*-heuristic algorithm for the given problem.

Therefore, it is clear that this study fills important gaps with attractive contributions that have never (or scarcely) been addressed in the published references.

3. Conceptual framework

Before developing our mathematical model for designing the OPTP, a conceptual framework for the problem is developed. The conceptual framework will aid in the definition of the optimization formulation. For this purpose, data pertaining to the tourism activities in the Sultanate of Oman were collected and analyzed. Oman is home to many cultural and eco attractions and, having the benefit of its geographical position, has a significant potential to become a major tourist destination in the region (Muthuraman & Al Haziazi, 2019). The data bank of the Ministry of Tourism is the major source of the analyzed data. Altogether, ten years of data were collected and these data were mainly related to the flow of tourists, their preferred mode of transportation, accommodation, touristic activities performed, budget, and total number of nights spent by tourists. The collected data helped in identifying the major trends in the Omani tourism industry.

From the analyzed data, it was found that the primary visitors to Oman are from the Gulf countries and the next from Europe. Even though most visitors are from the Gulf region, it was noticed that European visitors spend more nights in the country. This may be due to the major difference in culture between Oman and the European countries and also the distance separating the two regions. Moreover, other important information collected from the data analysis is summarized in Table 2. In the table, four different choices with the percentage of tourist preferences for each choice are presented. All the percentages in the table are presented in the rounded figures.

From the above data analysis, it was evident that the tours' definition problem is of multi-objective nature. Some tourists prefer minimizing costs of activities, but at the same time, maximizing the number of activities. Others may consider minimizing travelling time and/or distance. The data analysis suggested that a personalized approach for designing tourism packages is more appropriate than offering predefined packages, which is widely prevalent in the tourism industry. The

Table 2
Result of data analysis.

Item	1st choice (%)	2nd choice (%)	3rd choice (%)	4th choice (%)
Reason to visit Oman	Nature (38 %)	Culture (25 %)	Cheap destination (15 %)	Weather (12 %)
Accommodation	Hotel (42 %)	Rented house (28 %)	Apartment (18 %)	Hostel (6 %)
Activity of interest	Sea (42 %)	Desert (31 %)	Hiking (13 %)	City tour: museum, entertainment, etc. (8 %)
Duration of stay (Weeks)	less than1 (46 %)	1 (29 %)	1–2 weeks (12 %)	>2 (6 %)
Mode of transportation	Own vehicle (40 %)	Rented Vehicle (33 %)	Tour bus (25 %)	Others (2 %)
Locations visited per trip	2 (41 %)	1 (36 %)	3 (17 %)	>3 (6 %)
Planning criterion considered	Maximize # of activities (37 %)	Minimize activity cost (26 %)	Minimize travel time (21 %)	Minimize travel distance (12 %)

analysis helped in developing the conceptual framework, as shown in Fig. 1. The framework consists of a set of activities. In each activity category, there exists multiple options. For example, inside the heritage exploration category, the activities include forts, castles, traditional villages, etc. The tourists can select their preferred activities and their number is included in the pool of potential activities. Furthermore, they can input the total number of days they want to spend including their budget. To allow more flexibility, the budget is defined according to three thresholds—high, medium, and low. With these inputs, the mathematical model will identify the number of activities that the tourists can perform within the given constraints of time and budget and based on the objectives set by the tourist. Moreover, the model will help in identifying a day-to-day activities schedule and routes plan.

4. Mathematical model

The mathematical model is developed to optimize cost, distance travelled, and the number of activities. As the preferences of each tourist are different from the others, these multiple objectives or criteria are combined using a weighted sum. This helps each tourist to express the most suitable weights to each of the defined criteria based on his/her preferences. To develop the mathematical model, the following assumptions are considered:

- The trip can be planned along a multi-day horizon with T days.
- Each daily trip will start and end from the same accommodation.
- All tourists going for a given activity should leave that activity after spending a certain amount of time.
- The tourist may define a budget limit to control the total cost of the selected activities.
- Only one activity can be performed at any given time period and tourists should perform at least one activity per day.

In the model, the set of activities is M and each activity may belong to the sea category, sightseeing or hiking, etc. Among the elements of M, we define Q that represents the set of must-go activities. The inputs of the mathematical model are expressed in terms of subsets of j that are preferred by the tourist and the parameters that define the desires of the tourist. The notations used in the development of the mathematical model are illustrated in Table 3.

The full space OPTP model can be expressed as follows:

$$\textbf{Maximize} score = w_A f_1 + w_C f_2 + w_D f_3 \quad (1)$$

where:

Table 3
Notation used in the mathematical model.

Indices:	
i, j, r	Indices for the activities belonging to set M ({0} indicates the set of possible tourist's accommodations and Q indicates the set of must-go activities)
t	Indexing the day when an activity is conducted ($t = 1, 2, \ldots\ldots, T$)
Input:	
α_j	Expected time in hours to spend enjoying activity j
c_j	Cost to conduct activity j
d_{ij}	Traversal distance between activity i and activity j
B	Total budget limit set by the tourist
K	Time limit per day dedicated for activities
w_A	Weight assigned for the activities number
w_C	Weight assigned for the cost
w_D	Weight assigned for the travel distance
Max/Min (C)	Cost incurred if all the activities are (or if only one activity is) performed
Max/Min (D)	Distance travelled if all the activities are (or if only one activity is) performed
Max/Min (A)	Total number of available activities (=1 in the case of the *Min* operator)
Decision Variables:	
X_{ijt}	1, if the tourist travels from activity i to activity j during day t. 0 otherwise
Z_{jt}	1, if the tourist includes activity j in the itinerary of day t. 0 otherwise

$$f_1 = \frac{\sum_{j=1}^{|M|}\sum_{t=1}^{T} Z_{jt} - \min(A)}{\max(A) - 1} \quad (2)$$

$$f_2 = \frac{\max(C) - \sum_{j=1}^{|M|}\sum_{t=1}^{T} c_j Z_{jt}}{\max(C) - \min(C)} \quad (3)$$

$$f_3 = \frac{\max(D) - \sum_{i=1}^{|M|}\sum_{j=1}^{|M|}\sum_{t=1}^{T} d_{ij} X_{ijt}}{\max(D) - \min(D)} \quad (4)$$

Subject to:

$$\sum_{j\in M} X_{0jt} = 1 \forall t = 1, 2, \cdots.., T \quad (5)$$

$$\sum_{i\in M} X_{i0t} = 1 \forall t = 1, 2, \cdots.., T \quad (6)$$

$$\sum_{j\in M} \alpha_j Z_{jt} \leq K \forall t = 1, 2, \cdots.., T$$

$$\sum_{t=1}^{T}\sum_{j\in M} c_j Z_{jt} \leq B \quad (8)$$

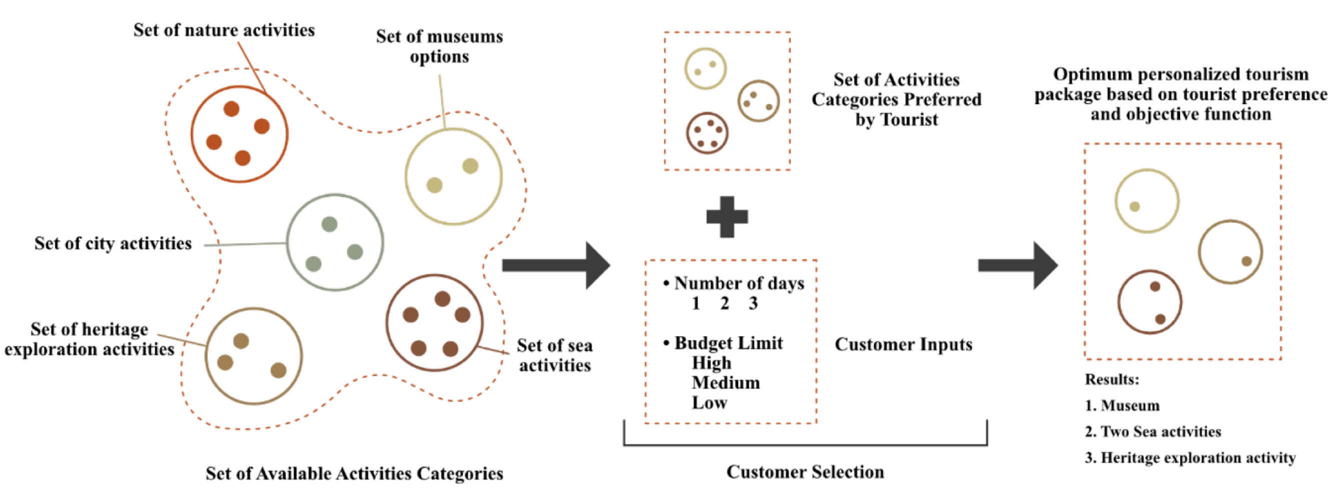

Fig. 1. Conceptual framework of our OPTP.

$$X_{0jt} + \sum_{i \in M \setminus \{j\}} X_{ijt} = X_{j0t} + \sum_{r \in M \setminus \{j\}} X_{jrt} \forall j \in M; \forall t = 1, 2, \cdots.., T \tag{9}$$

$$\sum_{t=1}^{T} Z_{jt} \leq 1 \forall j \in M \tag{10}$$

$$X_{0jt} + \sum_{i \in M \setminus \{j\}} X_{ijt} = Z_{jt} \forall j \in M; j \neq i; \forall t = 1, 2, \cdots, T \tag{11}$$

$$\sum_{t=1}^{T} Z_{jt} = 1 \forall j \in Q \tag{12}$$

$$\sum_{i,j \in S \cup \{0\}} X_{ijt} \leq |S| - 1 \forall S \subset M \cup \{0\}; \forall t = 1, 2, \cdots\cdots, T \tag{13}$$

$$X_{ijt} \in \{0, 1\} \forall i, j \in M \cup \{0\}; \forall t = 1, 2, \cdots\cdots, T \tag{14}$$

$$Z_{jt} \in \{0, 1\} \forall j \in M; \forall t = 1, 2, \cdots\cdots, T \tag{15}$$

The objective of the OPTP is to jointly optimize the cost, distance, and number of activities. Out of these three criteria, cost and distance criteria need to be minimized while the activity number criterion needs to be maximized. Moreover, each of these criteria has a different unit of measure. Therefore, in order to simultaneously optimize the three criteria, the normalized values of the criteria were integrated within the same objective function called *score*, as shown in Equation (1) (Piya et al., 2009). Thus, the objective here is to maximize the normalized weighted sum of the three criteria. Equation (2) normalizes the number of activities, which is to be maximized. On the other hand, Equations (3) and (4) normalize the cost and distance respectively and both the criteria need to be minimized. The normalization process is carried out as shown in Fig. 2. Fig. 2 (a) shows that for the criteria "activity", the normalized value increases as the total number of activities to be performed by the tourist approaches towards the total available activities. However, the normalized value decreases if the activity approaches towards its minimum, i.e., 1. Conversely, the opposite situation can be noticed for "cost" and "distance". As shown in Fig. 2 (b), the normalized value for these criteria increases as the criteria value approaches towards the minimum and vice-versa. The objective function will have a value of 1 if the maximum satisfaction of the weighted sum objective functions is reached and a value of 0 if the satisfaction is at its lowest score.

Constraints (5) and (6) are logical constraints that force the model to start the trip from one of the tourist's accommodation and end the trip at the same accommodation. Equation (7) states that the total hours spent on each daily trip shall not exceed a defined time limit. Equation (8) is the budget limit constraint. Equation (9) is a path continuity constraint, which states that every tourist entering activity j from i should leave j for another activity r or for his/her accommodation after spending some time. Equation (10) states that an activity can be conducted at most once during all the trip days. Equation (11) states that a route from the activity's location i to location j (including accommodation points) should exist only if both activities are conducted on the same day t. Equation (12) is related to the predefined set of must-go activities. This constraint will ensure that the tourist does not miss the most important attractions of the city (defined as the must-go set Q). Constraint (13) is the well known sub-tour elimination constraint (see for example Triki et al., 2017). Finally, constraints (14) and (15) define the domain of the binary decision variables X_{ijt} and Z_{jt}, respectively.

The developed model (1)–(15) results to be a combinatorial optimization problem whose size depends on the number of activities to be considered as well as on the extent of the planning horizon of the problem.

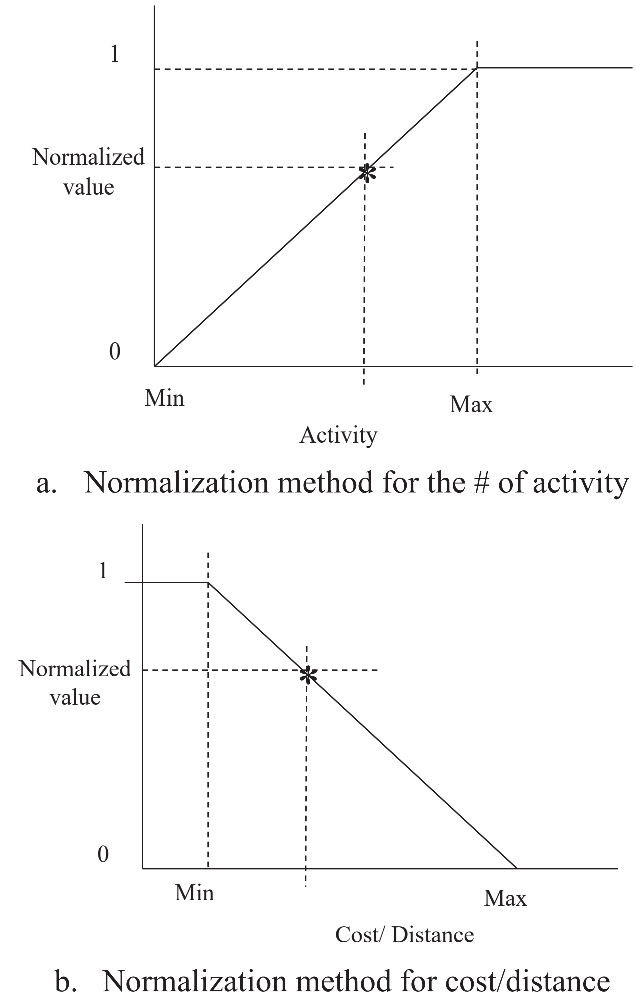

Fig. 2. Normalization for the maximization and minimization of criteria.

5. Solution methods

The optimization model dealt with in this research involves a routing problem, which is known to be NP-hard (Triki et al., 2020). It is expected that the developed full space mathematical model cannot solve the problem of large-scale instances related to bigger countries or cities within a reasonable computational time. Therefore, four *meta*-heuristic algorithms, namely NSGA-II, NSGA-III, MOGWO, and MOICA are developed to solve the proposed OPTP. In the following section, first, encoding and decoding procedure of a solution to the problem is explained. Thereafter, the method to improve a pool of randomly generated initial solutions for the algorithm is explained.

5.1. Solution representation

The problem presented in Section 4 involves sequencing a tourist's activities over a period of time (e.g., some days). The solution representation of the problem allocates each activity to a specific day and sequences the activities according to the day. NSGA-II and NSGA-III can work with both integer and real values for a solution. However, MOGWO and MOICA are continuous bases, i.e., they work with real values, so they require a continuous base solution representation. A single dimension array with a length equal to the number of activities plus the number of scheduling periods is proposed to represent a solution. This array contains real values between 0 and 1. Whenever a solution is generated (using algorithms or randomly), the value of problem variables and objective functions should be calculated under all constraints. The first step is to sort the solution and save the index. A day is represented by any value greater than the number of activities. Activities that occur before a particular day in this array are assigned to that day. These activities represent the sequence of activities for the day. Thus, this array represents the initial sequence of activities and the allocated day for each activity. Activities that occur after the last day will not be visited.

The constraints of the problem are then checked from day one based on this array. If the total time of activities for a day is violated, the last set of activities of that day are moved to the start of the next until the constraint is respected. Then the budget constraint is checked and if it is violated, an activity from the day with the most number of activities is removed until the constraint is respected. Finally, the algorithm will check for at least one activity for each day. If there is a day with no activity, then an activity from the day with the most number of activities is removed until the constraint is respected. Since all constraints are respected and a feasible solution is constructed, the objective functions are calculated and returned as the fitnesses of the algorithms.

For example, an array with the length of 15 is created for a tourist package with 10 activities for 5 days. The example array is [0.9, 0.13, 0.93, 0.86, 0.33, 0.08, 0.26, 0.87, 0.18, 0.93, 0.15, 0.96, 0.39, 0.67, 0.67]. The sorted indices are [6, 2, 11, 9, 7, 5, 13, 14, 15, 4, 8, 1, 3, 10, 12]. Therefore, activities 6 and 2 are allocated to day 1. Activities 9, 7, and 5 are allocated to day 3. Currently, no activity is allocated to days 4 and 5. Activities 4, 8, 1, 3, and 10 are allocated to day 2. Assuming that budget and time constraints are respected, the solution is corrected by removing the last activity, i.e., activity 10 from day 2 and allocating it to day 4. Once again, the solution is corrected by removing the last activity, i.e., activity 3 from day 2 and allocating it to day 5.

5.2. NSGA-II

NSGA-II is a multi-objective evolutionary algorithm having three characteristics, i.e., an approach for sorting nondominated solution, estimation of crowded distance, and operator for simple crowded comparison (Deb et al., 2000). Flowchart of NSGA-II is shown in Fig. 3. NSGA-II starts with producing a random initial population. By using a traditional selection method, this algorithm chooses s solutions from the population to create a mating pool. In this process, according to nondomination sorting, the algorithm considers the nondomination level based on current population. Also, the crowding distance is defined as the distance among a solution and other solutions.

To determine the best solutions of each s tournament, NSGA-II applies the following criterion: In a tournament wherein there are two solutions with various nondomination levels, the solution with the lower level will be selected for mating pool. Otherwise, out of two solutions with the same nondomination levels, the solution with higher crowding distance will be selected. After creating mating pool, crossover and mutation operators are implemented (as in) to produce offspring population from mating pool solutions. The crossover operator performs with two factors, pc and Dc, which are the crossover probability and crossover distribution index, respectively. Similarly, pm and Dm are the same factors for mutation operator.

After generating offspring population, it is merged with the current population. Then the algorithm selects the best s solutions by using nondominated sorting and crowding distance approach. The solutions in the new population are sorted in a group based on their domination level as {F1, F2, …}. Then, solution selection for a new generation starts with F1 and it continues based on their levels until the size of the new population reaches *s* or higher. In the selection process, when the new generated population size exceeds s, the selection of last level Fl is not fully included in the population. The section process is based on Equation (16). In the equation, *r* is the remaining solutions.

$$r = s - |F_1 \cup \cdots \cup F_{l-1}| \quad (16)$$

In order to choose the remaining *r* solutions of Fl level, the algorithm applies a selection process that takes into account the crowding distance of the solutions in Fl. This process measures the solutions in Fl into functions to arrange their distances from high to low and selects the r solutions with the largest distance. This approach helps to boost the choice of various nondominated solutions to maintain the diversity of the new population. Once the termination criterion is reached, i.e, maximum number of iterations, the algorithm will finish its function and provide the Pareto solutions of the last generation.

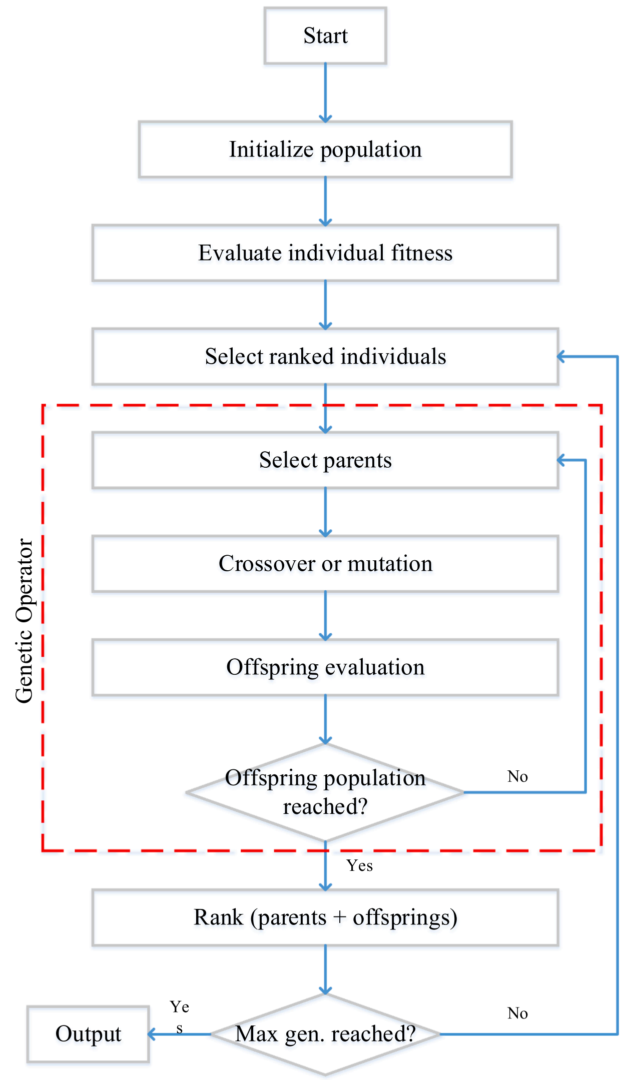

Fig. 3. NSGA-II flowchart.

5.3. NSGA-III

NSGA-III is developed to improve the functionality of NSGA-II when there are three or more objective functions (Deb & Jain, 2013). Fig. 4 illustrates the NSGA-III flowchart. NSGA-III uses a procedure similar to that applied by NSGA-II to generate an initial population. In this population, each solution demonstrates a feasible expansion. Then, NSGA-III uses the same crossover and mutation operators as NSGA-II to generate offspring population from the existing one. NSGA-III and NSGA-II differ in the selection process (SP) used to produce a new population with s solutions for the next generation by combining the current and offspring populations.

The NSGA-III selection procedure begins similarly to that of NSGA-II. The solutions are sorted into {F1, F2, …….}. Solution selection begins with F1 and continues until the size of the new population reaches s or

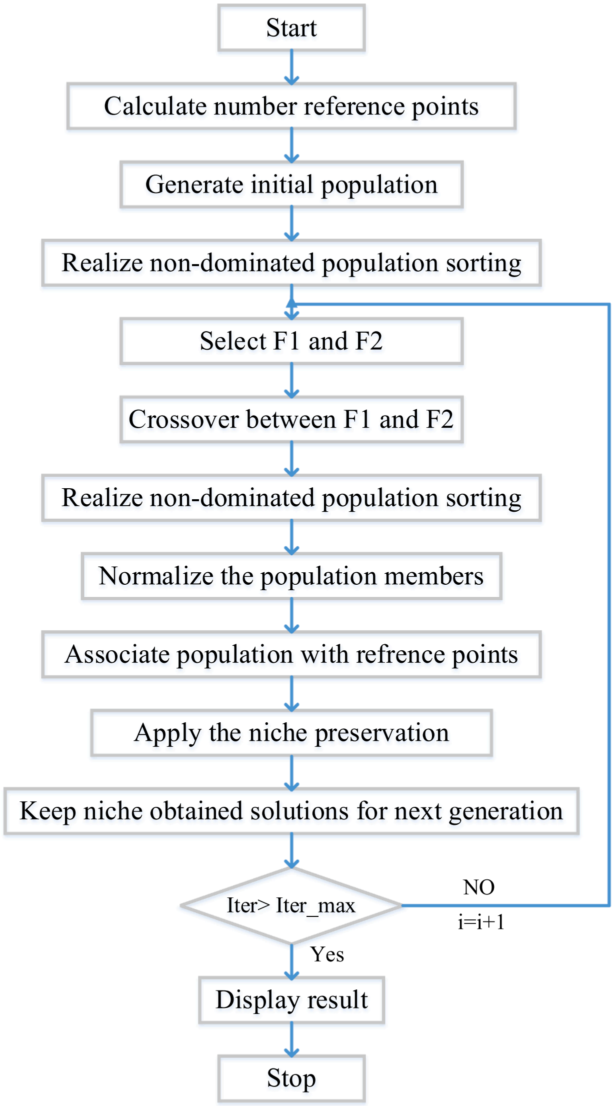

Fig. 4. NSGA-III flowchart.

higher. When the new population size trespasses s, NSGA-III applies an SP different from that used by NSGA-II to determine which r solutions from Fl to select. To choose the rest of the r solutions at the Fl level, NSGA-III uses a benchmark-based SP called reference points. This process includes a set of points extensively and evenly dispersed in the normalized hyperplane, which are the intrinsic reference points of the optimization problems solved by the algorithm. The process then underlines the choice of F1 solutions corresponding to the reference points. Therefore, this procedure enhances the choice of various and well-distributed nondominant solutions to maintain the variety and distribution of new populations. The termination process of NSGA-III shares the same criterion utilized by NSGA-II. After reaching this criterion, NSGA-III provides the Pareto set of the population corresponding to the last generation.

5.4. MOGWO

The Grey wolf optimization algorithm (GWO), developed by Mirjalili et al. (2014), is based on the social behavior and hunting mechanism of grey wolves. The grey wolf optimizer is motivated by the leadership level and hunting techniques of alpha (α), beta (β), delta (δ), and omega (ω) wolves. When GWO formulated the social hierarchy of wolves, it considered α wolf the most suitable solution. β and δ wolves are named the second- and third-best solutions. The rest of the potential solutions are supposed to be ω wolves. In this algorithm, the optimization is directed by α, β, and δ. To find the global optimum, the ω wolves follow these α, β, and δ wolves. The following equations inspire the surroundings of grey wolves during the hunt:

$$\vec{K} = \left|\vec{C}.\vec{Y_p}(t) - \vec{Y}(t)\right| \tag{17}$$

$$\vec{Y}(t+1) = \vec{Y_p}(t) - \vec{O}.\vec{K} \tag{18}$$

In Equations (17) and (18), t represents the present iteration, $\vec{O}$ and $\vec{C}$ are coefficient vectors, $\vec{Y_p}$ is the victim, and $\vec{Y}$ is prey wolf positions. $\vec{O}$ and $\vec{C}$ are measured based on the following formulation:

$$\vec{O} = 2\vec{o}.\vec{m}_1 - \vec{o} \tag{19}$$

$$\vec{C} = 2.\vec{m}_2 \tag{20}$$

The elements of o decline linearly from 2 to 0 in the iterative process, and m_1 and m_2 are distributed randomly within the range [0, 1]. To determine the optimal solution, the GWO algorithm applies its two main mechanisms. The algorithm stores the three best solutions and prompts other search agents to upgrade their positions relative to them. During the optimization process, the following formulae are run continuously for each search agent to simulate the search and find promising areas of the search space:

$$\vec{K}_\alpha = \left|\vec{C}_1.\vec{Y}_\alpha - \vec{Y}\right|; \vec{K}_\beta = \left|\vec{C}_2.\vec{Y}_\beta - \vec{Y}\right|; \vec{K}_\delta = \left|\vec{C}_3.\vec{Y}_\delta - \vec{Y}\right| \tag{21}$$

$$\vec{Y}_1 = \vec{Y}_\alpha - \vec{O}_1.\left(\vec{K}_\alpha\right); \vec{Y}_2 = \vec{Y}_\beta - \vec{O}_2.\left(\vec{K}_\beta\right); \vec{Y}_3 = \vec{Y}_\delta - \vec{O}_3.\left(\vec{K}_\delta\right) \tag{22}$$

$$\vec{Y}(t+1) = \frac{\vec{Y}_1 + \vec{Y}_2 + \vec{Y}_3}{3} \tag{23}$$

To solve multi-objective problems using GWO, two new elements were proposed (Mirjalili et al., 2016). One is responsible for organizing the currently obtained non-dominated Pareto optimal (PO) solution into an archive. The other is responsible for a leader selection strategy from non-dominated solutions. These elements help select solutions as leaders from the archive during hunting. The first element (archive) stores PO solutions. Three α, β, and δ wolves are then selected from the archive at each iteration. To find a solution near the global optimum, this selection is applied by a roulette-wheel approach with Equation (24). In the equation, the number of POs in section i is defined as N_i, and c is a fixed number greater than one.

$$P_i = \frac{c}{N_i} \tag{24}$$

The flowchart of MOGWO is as shown in Fig. 5. The GWO has been confirmed to need search agents to suddenly relocate in the early stages of optimization and progressively relocate in the final stages (Mirjalili et al., 2014). This performance assures that the GWO converges in the search area (Van den Bergh & Engelbrecht, 2006). The MOGWO algorithm contains all the attributes of GWO, so search agents similarly navigate and use the search area. The chief alteration is that MOGWO explores the archive member set, while GWO keeps and enhances only the three best solutions.

5.5. MOICA

MOICA integrates the contest between empires and applies the idea of empire colonization (Atashpaz-Gargari & Lucas, 2007). The basic notion of MOICA is that each empire has multiple non-dominant solution sets, namely imperialists, and one Global Non-dominated solution

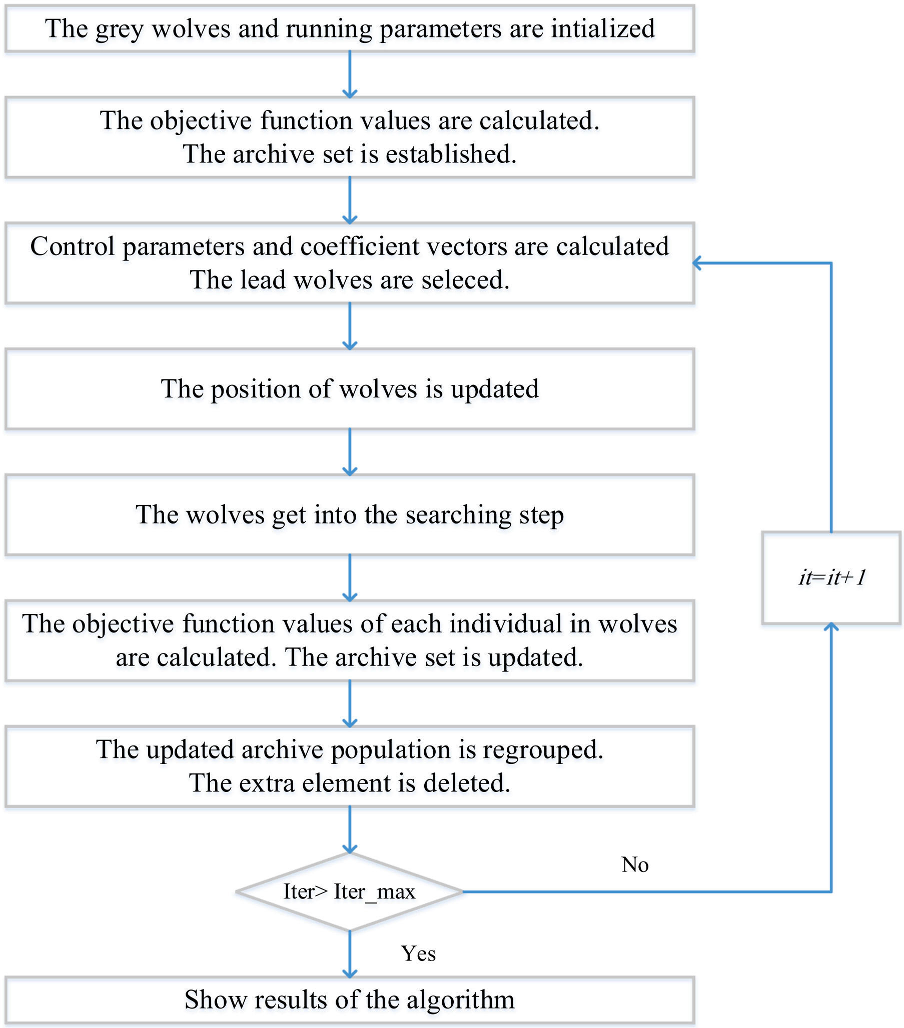

Fig. 5. MOGWO flowchart.

(GNDS) set that includes the best imperialists of all empires. Empires try to own the colonies of other nations due to their own power, which allows strong empires the chance to handle multiple colonies with lower power. In one iteration, the colonies of each empire change by changing their cost values with respect to their position. Thus, imperial colony, C, can be improved and become better than the current imperialist groups. In this situation, the new colony with a better cost turns into a member of the imperialists of the empire, that is, a member of the set of non-dominated solutions. Therefore, the former imperialism I, which was ruled by C, becomes a colony.

The execution of this algorithm has a simple characteristic. Primarily, N empires exist and every empire has a Pareto optimal solution, or local non-dominated solutions (LNDSs). The total number of LNDSs is generally N. A GNDS set is also acquired from N LNDSs. As each empire's LNDSs set is modified during each iteration, the GNDS is also modified accordingly. This algorithm has one GNDS for the entire execution, while the number of LNDSs steadily declines as the competitive middle empire collapses.

In MOICA, non-dominance is measured as follows. A solution with a value of 1 belongs to the first front, and a solution with a front value of 2 is an expression allocated to the second. Therefore, the LNDS and GNDS sets include only the solutions owned by the first front. For diversity preservation, this algorithm does not use special parameters, which is one of its significant characteristics, allowing it to avoid additional calculations to maintain the spread of solutions. Although MOICA does not use shared parameters, it has an excellent solution due to the results obtained in simulations and experiments by the assimilation method used in the algorithm. As defined earlier, all colonies of an empire move towards one available imperialist. Nevertheless, the imperial colony transfers to *I*, one of the imperialists, with the GNDS set. The imperialist, I, to which the colony transfers is chosen by chance at each iteration in the GNDS set. Therefore, the preclusion of shared parameters derives from the multi-objective nature of the algorithm; every solution in a non-dominated solution set is correct, and no single solution exists. The flowchart of this algorithm is presented in Fig. 6.

6. Numerical results

A numerical investigation is conducted to validate the mathematical model and to understand the effect of the various parameters on the model. First, a small real case example related to the city of Muscat is solved, followed by solving a greater realistic hypothetical problem. We then tune the parameters of the *meta*-heuristic algorithms, followed by a numerical analysis that proves the validity of the algorithms. Thereafter, the four proposed algorithms are compared to determine the best one for the problem under consideration. Moreover, a sensitivity analysis is also conducted to understand the effect of the weights of the objective functions on the total score. The full space mathematical model is implemented using the commercial LINGO programming language 18.0.44 (https://www.lindo.com) and solved by the optimization

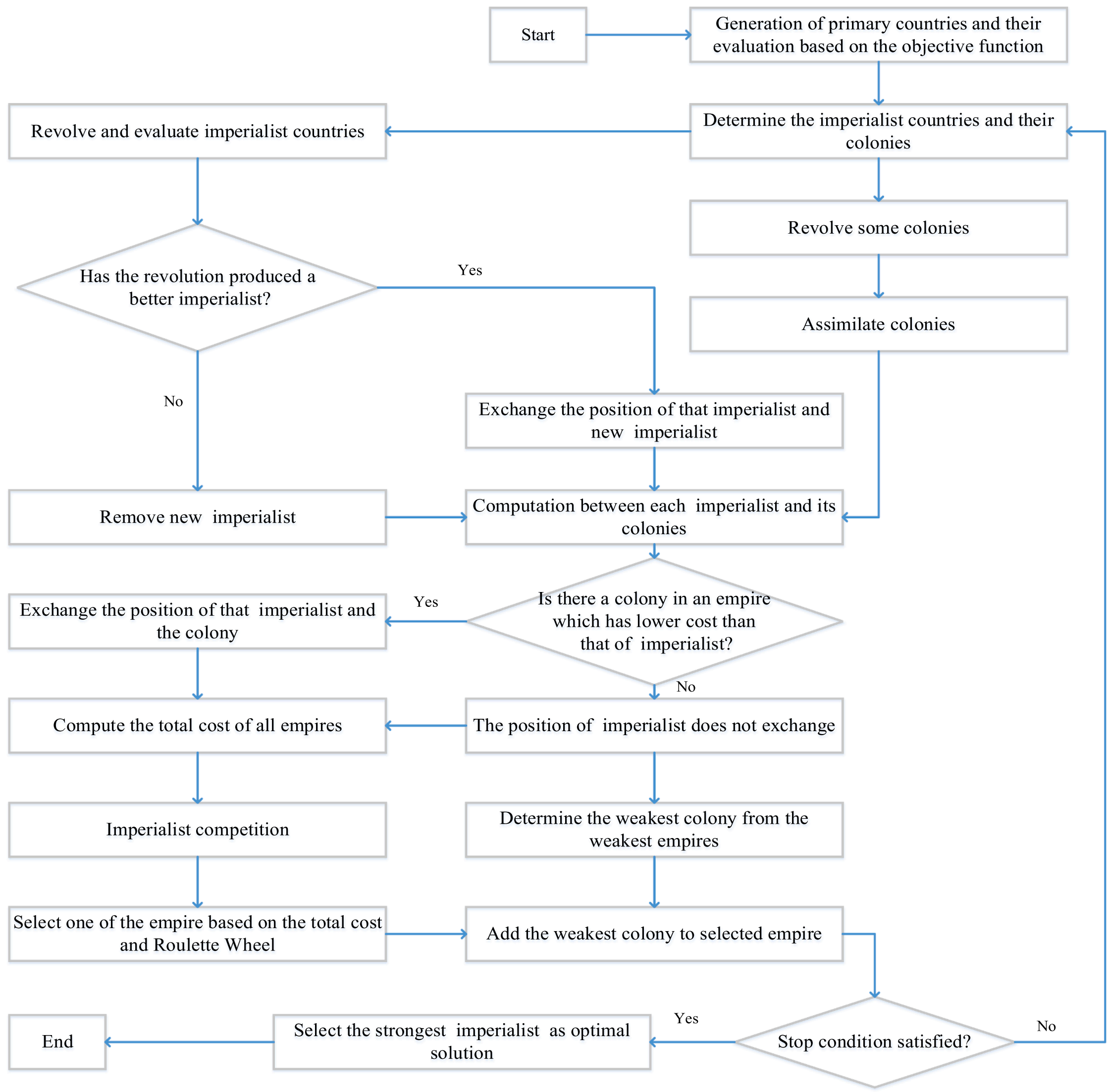

Fig. 6. MOICA flowchart.

packages it embeds. The *meta*-heuristic algorithms are developed using MATLAB version 2017b software. The test instances are executed on a computer having a Quad-Core Intel Core i5 processor with a CPU of 2.30 GHz and 8 GB of RAM.

6.1. Case example

The case example was conducted by communicating with a group of tourists that recently visited Oman and decided to spend two days exploring its capital city, Muscat. The input data revealed by them consists of the categories of activities of their interest, the budget limit, and the number of days for which they need personalized packages. They were mainly interested in the heritage activities, museums, and city tour with shopping. As it was their first visit to the country, they did not mention any must-go activities among their destinations. The group mentioned that the trip was limited to only two days and they were staying at a hotel near the area of their interest. Therefore, they were unwilling to change their accommodation during the days covered by the planning horizon. They requested to generate two alternative itineraries, one with a high budget and another with a medium one. With the above information, five locations for heritage activities (indexed as 1, 2,.., 5), three for shopping activities (indexed as 6, 7,.., 10), and five museum activities (indexed as 11, 12,…, 15) were considered. Further communication with them highlighted that the group prefers to organize as many activities as possible but considers minimizing the cost as a greater priority than minimizing the travel distance. This information helps in identifying the weights ($W_A = 0.5$; $W_C = 0.3$; $W_D = 0.2$) to be considered in the objective function.

The results obtained from the optimization solver are reported in Table 4. The row labeled "Itinerary" in the table shows the activities suggested to be performed on each day and their sequence along each day's route. For example, the scheduled activities on day 1 for the high budget scenario started with visiting one of the heritage locations,

Table 4
Result obtained for the case example.

Items	High Budget		Medium Budget	
	Day 1	*Day 2*	*Day 1*	*Day 2*
Itinerary	0-1-8-3-2-4-7-6-0	0-11-13-5-15-14-0	0-1-9-7-3-2-0	0-6-5-10-14-0
Total number of activities	7	5	5	4
Daily cost ($)	178.6	66.5	65	74.5
Total cost ($)	245.1		139.5	
Travel distance (km)	45	12	16.9	43.4
Score	**0.871**		**0.842**	

followed by a shopping center and so on and concluded with again shopping at location 6, before coming back to the accommodation. There were no museum visits on day 1. However, day 2 started with a visit to two museums and concluded by visiting the other two museums with a shopping break in the middle. On day 2, out of the 5 activities, four were related to museum visits. This was because most of the major museums in Muscat are concentrated in one specific zone of the city and far away from most other attractions. Therefore, scheduling them on the same day would reduce the overall travel distance. From the results of Table 4, it is clear that the number of activities performed with a high-budget scenario is higher compared to its medium-budget counterpart, even though the travel distance in the case of the medium-budget scenario is higher. Also worth noting is that the medium budget itinerary involves visiting only one museum on both days. This limitation is clearly due to the preference of the tourists for assigning a higher weight to the cost minimization in the objective function, given that usually museum entrance costs more. For both the budget scenarios, the objective function score, which represents the cumulative weighted level of satisfaction achieved with respect to the tourist preferences, is above 0.8, which is close to the maximum score i.e., 1.

6.2. Hypothetical test problem

Next, to understand the complexity level of our model, a further numerical experiment is conducted with a hypothetical instance. The experiment mainly aims to assess the effect of increasing the problem's size and measuring the problem's complexity. The experiment also allows measuring the robustness of the obtained solution by evaluating the impact of altering some input parameters of the model such as the number of activity categories, the budget level, and the trip duration. It should be noted that, as is evident from Fig. 1, increasing the number of categories will increase the number of total activities to be considered, as each category will involve an additional set of activities. For this hypothetical test problem, we consider 10 activities for incorporation into each category in the input data set. The set of instances considered in this experiment is reported in the first column of Table 5. The number of activity categories and visit days planned vary from [1–3] and [2–6], respectively. This means that the total number of activities ranges between 10 and 30. Moreover, three budget levels, i.e., high (H: $500), medium (M: $350), and low (L: $200) are considered for the analysis. With the combinations of activity category, planned days, and budget levels, altogether 27 combinations of the problem are generated and analyzed. For the analysis, the cost for each activity, distance from one activity to another, and activity time are randomly generated within the intervals [10–30], [8–12], and [1–5], respectively. Moreover, equal weights are considered for three criteria in the objective function. The table shows the model's solution in terms of the selected number of activities, total cost, total distance, score, and CPU timing needed to execute the model by the Lingo solver.

The results show that the number of activities, associated total cost, and traveled distance are affected by the combination of all three parameters considered in the experiment. Generally, with the increase in the value of these parameters, the number of selected activities increases. This increase is because increasing the number of planning days will increase the time limit, allowing the inclusion of more activities within the multi-day trip. Further, increasing the budget limit will allow the inclusion of more activities in the trip plan. However, the result is not true when there are constraints on available activity, budget, and limits on the planning day. For example, when the budget is low, then the number of selected activities is in the range of (8–11) irrespective of an increase in the number of available activities and planning day. This range is because the total cost of the selected activities approaches closer to the budget limit. The maximum number of selected activities reaches 27 when the number of available activities, planning days, and the budget limit is at the highest level. In most of the setup combinations, cost and distance increase with the increase in the number of selected activities, with few exceptions (as, for example, experiments 13 and 20). One reason for the exceptional cases is that the selected activities take less cost to perform. Moreover, the exception may also be due to the selected activities being comparatively closer so that the total distance covered is lower.

The budget limit appears to significantly constrain the solution when high-cost activities were selected as part of the model input. The lower the budget level, the fewer the activities conducted; hence, less distance

Table 5
Result obtained for the hypothetical test problem.

#no	Number of Activities	Planning days	Budget	No of Selected Activities	Cost ($)	Distance (km)	Score	CPU time (min: sec)
1	10	2	L: 200	8	172	89	0.80	00:01
2			M: 350	10	183	85	0.91	00:15
3			H: 500	10	183	85	0.96	01:36
4		4	L: 200	10	169	91	0.82	00:32
5			M: 350	10	169	91	0.92	00:58
6			H: 500	10	169	91	0.94	01:07
7		6	L: 200	10	169	91	0.82	03:11
8			M: 350	10	169	91	0.92	01:09
9			H: 500	10	169	91	0.97	00:19
10	20	2	L: 200	8	172	89	0.80	00:00
11			M: 350	11	199	95	0.77	00:01
12			H: 500	12	206	96	0.79	00:12
13		4	L: 200	10	185	93	0.77	01:06
14			M: 350	19	327	162	0.77	02:00
15			H: 500	20	387	176	0.79	16:00
16		6	L: 200	10	178	86	0.84	06:00
17			M: 350	20	331	178	0.77	07:04
18			H: 500	20	387	176	0.79	10:40
19	30	2	L: 200	10	182	76	0.73	00:01
20			M: 350	11	174	88	0.81	00:11
21			H: 500	12	198	103	0.76	00:12
22		4	L: 200	11	189	84	0.73	00:01
23			M: 350	19	324	159	0.75	02:03
24			H: 500	20	387	176	0.79	10:48
25		6	L: 200	11	194	85	0.73	13:01
26			M: 350	20	324	164	0.75	14:16
27			H: 500	27	417	233	0.76	12:12

is traveled. For all the experiments, the score is above 0.7 with an average score of 0.81 over all the instances, signifying that the model can generate the solution with a high customer satisfaction level. Table 5 shows that the computational time increases with the increase in the number of activities and planning days. When the planning horizon is two days, the computational time is almost negligible. However, when the number of activities is 30 and planning days are 6, then the computational time reaches>13 min. This change is because with the increase in the number of activities and days, the number of possible combinations of activities to be selected and scheduled within the given planning days increases remarkably.

6.3. Algorithms parameters tuning and validation

Before evaluating the algorithms and choosing the best, it is necessary to tune their parameters as the performance of *meta*-heuristics is highly impacted by the parameters. The experiment designed using the Taguchi method (Taguchi 1986) has been extensively used for this purpose (Mokhtarzadeh et al., 2021). For each algorithm, a three-level Taguchi design is used to design the experiment. The average score (Equation (1)) of the solutions provided by each algorithm is considered the response level. A medium-size instance with 100 activities and 14 days is considered during the experiment.

Table 6 shows the parameters tuned for the proposed algorithms and the best values obtained for these parameters. For NSGA-II, nine experiments were conducted, while for other algorithms, twenty-seven were conducted, as the number of parameters to be tuned for these algorithms was more compared to that for NSGA-II. The results were analyzed using Minitab software to identify the best values. As a sample, the result generated by Minitab software for NSGA-II is as shown in Fig. 7.

To validate the performance of *meta*-heuristic algorithms, their solutions are then compared with the exact method on nine small-size instances. The results of these instances are as shown in Table 7. From the result, the gap between the exact method solutions and the four algorithms, NSGA-II, NSGA-III, MOGWO, and MOICA, is clearly less than 5 %. This gap implies that the algorithms are reliable and can be used to solve large-size instances.

6.4. Algorithms performance comparison

To compare the performance of solution methods, 36 test instances were randomly generated. The number of activities ranges from 10 to 200, and the number of days ranges from 2 to 30. The expected time for each activity, cost to conduct the activity, and traversal distance between the activities were randomly generated between [1, 5], [10, 30], and [8, 12], respectively. The time limit per day dedicated to the activities was 8. The 36 generated instances were solved by the four proposed algorithms, and their performances were analyzed based on six metrices.

Table 6
Parameter tuned and the best values for different algorithms.

Parameter tuned	NSGA II	NSGA III	MOGWO	MOICA
No of population	100	100	75	–
No of iteration	200	150	200	–
% of crossover children	0.8	0.9	–	–
% of mutation children	0.1	0.3	–	–
Number of divison	–	20	–	–
Mutation rate	–	0.2	–	–
Leader selection pressure	–	6	4	–
Archive size	–	–	40	–
Grid inflation parameter	–	–	0.15	–
Number of grids per dimension	–	–	20	–
Archive member selection pressure	–	–	2	–
Number of initial countries	–	–	–	160
Number of Initial Imperialists	–	–	–	40
Number of decades	–	–	–	200
Revolution rate	–	–	–	0.2
Uniting threshold	–	–	–	0.02
Maximum percent of Imperialists that can be contained in the Empire	–	–	–	0.4

6.4.1. Quality metric

When the results from all algorithms are combined, the quality metric for an algorithm is the proportion of the Pareto front it produces (Rabbani et al., 2018). A higher quality metric represents better algorithm performance. The results obtained for the quality metric are as shown in Fig. 8. MOGWO's quality metric is zero for all 36 test instances, meaning that all MOGWO''s solutions are dominated by other algorithms and that the algorithm is unable to produce high-quality solutions. MOICA produced an average of 17 % of Pareto front solutions. The proportion of MOICA's solutions in the set of non-dominated solutions decreases as problem size increases. NSGA-III produced fewer non-dominated solutions than NSGA-II did, despite being an updated version of NSGA-II. On average, NSGA-II produced 55 % of Pareto front solutions, while NSGA-III produced 27 %. Additionally, the portion of the NSGA-III (NSGA-II) algorithm's solutions in the set of non-dominated solutions increases (decreases) with problem size. In large instances, NSGA-III outperformed NSGA-II. Finally, a two-sample *t*-test is conducted to determine which algorithm is superior. NSGA-II and NSGA-III perform equally in terms of the quality metric. NSGA-II and NSGA-III are both superior to MOGWO and MOICA. Therefore, they can be used to obtain high-quality solutions.

6.4.2. Mean ideal distance

The mean ideal distance is calculated as the sum of the distance of all solutions, in the Pareto front of algorithms, from the ideal solution (Karimi et al., 2010). An ideal solution consists of the maximum value of the objective function obtained, concerning all constraints, from the algorithms. Therefore, an algorithm with less of this criterion is more efficient. Fig. 9 depicts the result for the mean ideal distance metric. NSGA-II, NSGA-III, and MOICA have average ideal distance metrics of 0.68, 0.64, and 0.67, respectively. The Pareto fronts of these three algorithms have approximately equal distances from the ideal solution. Although the quality metric shows that only about 17 % of MOICA solutions are among the non-dominated solutions from all algorithms, but the mean ideal distance reveals that MOICA solutions are near the Pareto front. Therefore, MOICA solutions are not low quality. Furthermore, a two-sample *t*-test shows that the three algorithms perform equally on average in terms of the ideal distance metric. However, MOGWO could not provide high-quality solutions.

6.4.3. Spacing metric

The spacing metric indicates how the dominated solutions are distributed. algorithms with a lower spacing metric perform better (Rabbani et al., 2020). The results obtained for the spacing metric are as shown in Fig. 10. The averages of the spacing metric over 36 instances for NSGA-II, NSGA-III, MOGWO, and MOICA are 0.153, 0.0612, 0.077, and 0.1, respectively. A two-sample *t*-test confirmed the superiority of NSGA-III and MOGWO over other algorithms. However, as the MOGWO solutions are not as good as NSGA-III solutions in terms of the quality metric. We select NSGA-III as the superior algorithm in terms of the spacing metric.

6.4.4. Diversity metric

The diversity metric is measured as the distance between the non-dominated solutions of an algorithm. The algorithm explores a larger space of solutions when this metric has a high value. Therefore, the higher the value, the better this metric (Rabbani et al., 2020). The results obtained for the diversity metric are as shown in Fig. 11 and indicate the superiority of NSGA-III over other algorithms. The average NSGA-III diversity metric value is 360, while the average of MOGWO, MOICA,

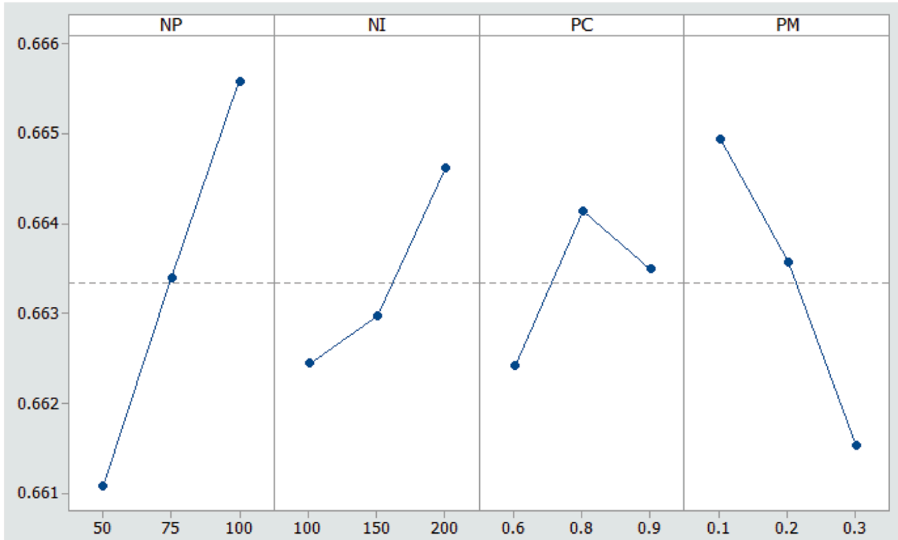

Fig. 7. NSGA-II parameters tuning analysis diagram.

Table 7
Meta-heuristics performance validation results.

# no	No. of Activities	Planning Days	Budget	Exact method (Score)	NSGA-II		NSGA-III		MOGWO		MOICA	
					Score	Gap (%)	Score	Gap (%)	Score	Gap (%)	Score	Gap (%)
1	10	2	200	0.80	0.76	5	0.77	4	0.75	6	0.78	3
2		4	350	0.92	0.88	4	0.9	2	0.89	3	0.86	7
3		6	500	0.97	0.96	1	0.97	0	0.93	4	0.92	5
4	20	2	200	0.80	0.77	4	0.79	1	0.76	5	0.74	8
5		4	350	0.77	0.72	6	0.73	5	0.71	8	0.73	5
6		6	500	0.79	0.76	4	0.78	1	0.75	5	0.73	8
7	30	2	200	0.73	0.71	3	0.71	3	0.7	4	0.69	5
8		4	350	0.75	0.74	1	0.7	7	0.73	3	0.7	7
9		6	500	0.76	0.71	7	0.72	5	0.69	9	0.7	8

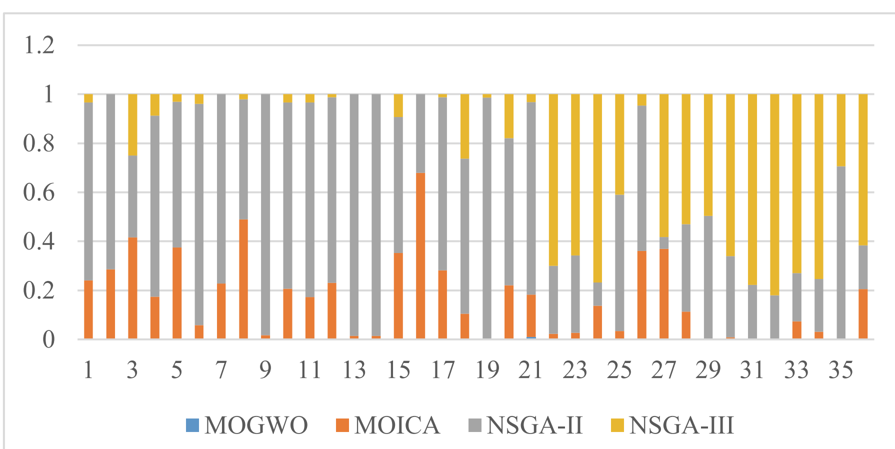

Fig. 8. Performance measure in term of quality metric.

and NSGA-II diversity metric values are 10, 111, and 99, respectively. Moreover, a two-sample *t*-test proved the superiority of NSGA-III over others.

6.4.5. Number of Pareto solutions

Fig. 12 illustrates the number of Pareto solutions for the algorithms. On average, NSGA-II found 61 non-dominated solutions, which is much more than other algorithms. NSGA-III follows with an average of 36 solutions. A two-sample *t*-test proved that NSGA-II was superior. NSGA-II allows a decision maker to look at different solutions. However, the spacing and diversity metrics indicate that NSGA-III explores the solution space more efficiently. In other words, NSGA-III provides a wider range of solutions. Therefore, although NSGA-II is better in terms of the number of Pareto solutions metrics, we suggest using NSGA-III.

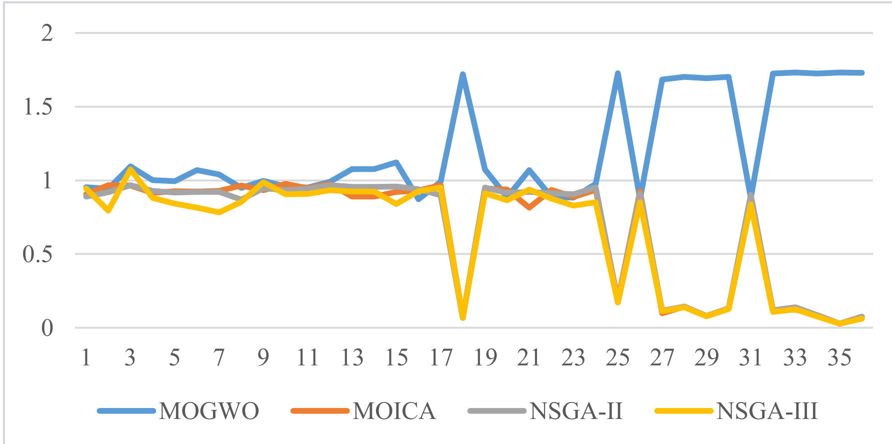

Fig. 9. Performance measure in term of distance from ideal point.

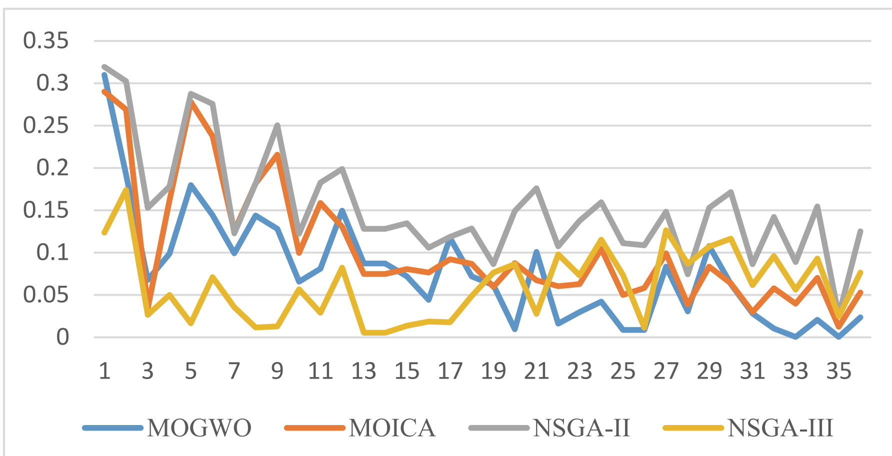

Fig. 10. Performance measure in term of spacing metric.

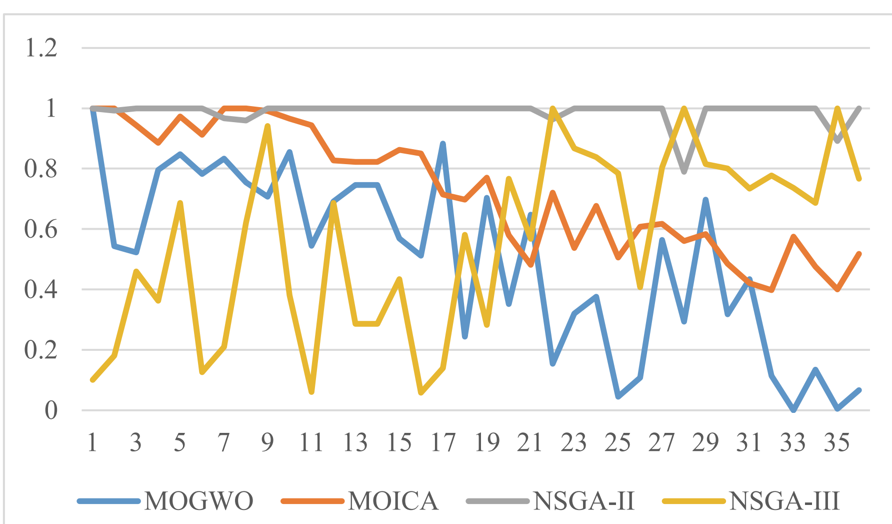

Fig. 11. Performance measure in term of diversity metric.

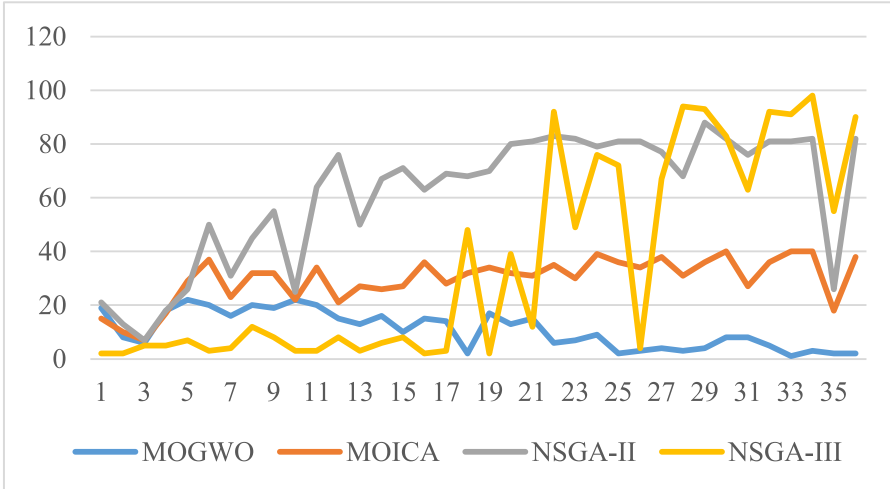

Fig. 12. Performance measure in term of number of Pareto solutions.

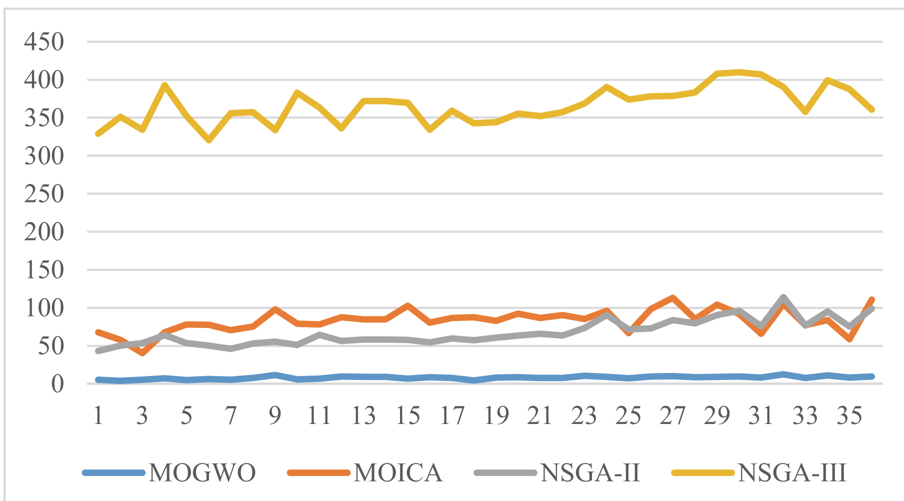

Fig. 13. Performance measure in term of solution time (sec).

6.4.6. *Solution time*

Fig. 13 compares the algorithms in terms of the time required to execute the problem. Results indicate that the NSGA-III solution time is higher than others', while the time difference between solution times of NSGA-II, MOICA, and MOGWO is not significant.

In summary, comparing MOGWO, MOICA, NSGA-II, and NSGA-III reveals that NSGA-II and NSGA-III provide high-quality solutions, while MOGWO's solutions are of lower quality. Furthermore, the spacing and diversity metrics demonstrate that NSGA-III is better at exploring solution space than NSGA-II is. However, NSGA-II is better in terms of the number of Pareto solutions and solution time. Thus, when the solution time is of great importance, NSGA-II is a better choice. When a diverse range of solutions is important, NSGA-III is a good choice.

6.5. *Comparison of NSGA-II and NSGA III with genetic algorithm*

To understand the need of optimizing multiple objectives for the tourism itinerary problem and to validate the results of our best algorithms, further experiments were conducted using a genetic algorithm (GA). GA is designed to optimize only a single objective at a time, i.e., either cost or distance, or time. The results of GA are then compared with those obtained from the best heuristic algorithms for the multi-objective optimization problem. The single-objective version of the problem is designed in two ways, namely GA-I and GA-II.

- *GA-I:* At a given problem instance, the GA is run to optimize each objective function separately (single-objective version), and the optimal results of the objective functions (cost, distance, and number of activities) were combined together to obtain a score.
- *GA-II:* The GA is run to optimize any one of the objective functions randomly. At the optimal value of the selected objective function, we identified the results for the other two objective functions (e.g., if we optimize cost, then at the optimal cost we identify the distance and number of activities). Then, the results of these objective functions were combined to convert them into a score.

Table 8 compares the GA results with those of NSGA-II and NSGA-III. The results show that for most of the problem instances, the results of NSGA-II and NSGA-III are better than those of GA-I and GA-II. In terms of the average percentage gap, the results of GA-I are, in fact, very close to those of NSGA-II and NSGA-III. The result is slightly better than that of NSGA-II but slightly worse than that of NSGA-III. However, NSGA-II and NSGA-III outperform GA-II. On average, the performance of NSGA-II and

Table 8
Comparison of the results of NSGA-II & NSGA-III with GA.

# no	No. of Activities	Planning Days	Budget	GA-I			GA-II		
				Score	Gap		Score	Gap	
					NSGA-II	NSGA-III		NSGA-II	NSGA-III
1		2	200	0.77	−1.32	0.00	0.68	10.53	11.69
2	10	4	350	0.87	1.14	3.33	0.85	3.41	5.56
3		6	500	0.91	5.21	6.19	0.90	6.25	7.22
4		2	200	0.77	0.00	2.53	0.72	6.49	8.86
5	20	4	350	0.75	−4.17	−2.74	0.68	5.56	6.85
6		6	500	0.76	0.00	2.56	0.66	13.16	15.38
7		2	200	0.7	1.41	1.41	0.61	14.08	14.08
8	30	4	350	0.73	1.35	−4.29	0.68	8.11	2.86
9		6	500	0.74	−4.23	−2.78	0.65	8.45	9.72
Average gap (%)					−0.60	0.69		8.45	9.14

NSGA-III is around 8 % and 9 % better than that of GA-II, respectively. The result demonstrates the quality of the proposed algorithms and the need to develop a model for multi-objective optimization for the tourist itinerary problem.

6.6. Sensitivity analysis and managerial insights

As the weights of the criteria will significantly impact our mathematical model outcomes, a sensitivity analysis is conducted to know the effect of changing the weight of the objective functions. For the sensitivity analysis, the weight of each criterion is randomly varied into three levels (0, 0.333, and 0.667). For example, if the weight of the criterion "number of activities" is zero, then the weight of "cost" will be either 0.333 or 0.667 and that of "distance" will be either 0.667 or 0.333, such that the sum of the three weights is always 1. The effect of changes in the weights is analyzed based on the main effect plots, and the analysis is carried out using the Minitab software at a 95 % confidence interval. The plots are used to observe the effect on the number of activities considered, total cost, distance traveled, and score with respect to the change in the weights assigned to the objective functions.

Fig. 14 shows the main effect plot for several activities when the weights of activity, cost, and distance in the objective function change. The plot shows that as the weight given for the number of activities increases, the latter increases and the effect is remarkable. Even though the number of activities increases with the increase in the weight for the distance, the effect is moderate. In contrast, increasing the weight of the cost function has the opposite effect on the number of activities, and the effect is moderate. Similarly, we carried out several additional experiments to analyze the combined effect of different weights of the total cost, distance traveled, and score. For the sake of brevity, not all the plots were reported here, rather the results obtained from the analysis are summarized in Table 9. From the table, the total cost evidently increases with the increase in the weight of activity and distance functions. However, the effects are high and moderate, respectively. Also of note is that the total cost decreases with the increase in the weight for the cost function and that the effect is significant. The distance traveled

Table 9
Summary of the main effect plots.

Effect on:	Weight increase of:		
	Activities	Cost	Distance
Number of Activities	Increase (High)	Decrease (Moderate)	Increase (Moderate)
Total Cost	Increase (High)	Decrease (High)	Increase (Moderate)
Distance traveled	Decrease (High)	Decrease (High)	Decrease (High)
Score (objective value)	Increase (High)	Increase (Moderate)	Increase (High)

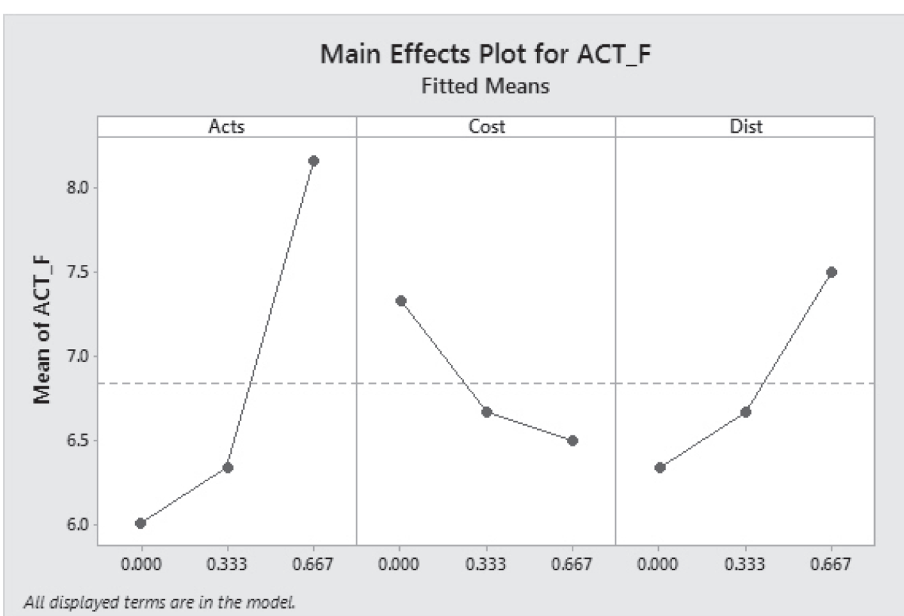

Fig. 14. Main effect plot on activities function.

decreases with the increase in the weights of all the functions, and even here, the effect is notable. On the other hand, the score increases as expected with the increase in the weights of all the functions. The effects are high for the activities' number and distance while it is moderate for the cost weight.

The above sensitivity analysis results provide very valuable managerial insights to the tourism service provider, who will benefit from the observed trends in increasing customer satisfaction. For example, the service provider should discourage his/her customers from any weight combination that may raise the cost component. This is because usually customers are sensitive to the cost component and will feel unfulfilled or frustrated with significant cost increases (even though still within their pre-specified budget threshold) as a result of a minor change in some of the criteria weights. Usually, sales managers are perfectly aware of these kinds of techniques but need to be supported by quantitative insights, such as those reported in Table 6 to ensure that customers are satisfied with the touristic package they are selecting, even compared with other available and feasible ones.

7. Conclusion

The tourism supply chain aims to satisfy the needs of the end-users based on their preferences. However, the preferences of each tourist may differ from others. Some tourists would prefer cost reduction, while others would be satisfied if they could enjoy more activities within the given planning period and preset budget limit. Moreover, tourists may have several preferences and may want to concurrently optimize multiple criteria. Consequently, from the optimization point of view, tourist preferences can be expressed through a multi-objective framework, such that some objectives need to be minimized while others need maximization.

This research aims to design optimum personalized tourism packages based on the preferences of the tourists. The research developed a mathematical model that optimizes multiple objectives using the weighted average method based on tourists' preferences when they plan a trip. The method allows tourists to express their preferences for one objective over others by assigning the most appropriate weight to each criterion. At first, the mathematical model was tested on a small case example to check its validity and applicability. The model was able to generate meaningful itineraries for the two-day planning horizon with the activity sequence for each day under high- and medium-budget constraints. Further, the model was tested in a realistic hypothetical test setting for an extended planning horizon and with a higher number of activities. For all the test instances, the score is above 0.7, indicating that the model can generate touristic plans with a high satisfaction level. As the problem under study is NP-hard, *meta*-heuristic approaches such as NSGA-II, NSGA-III, MOGWO, and MOICA were proposed to solve large-scale problems. The numerical analysis shows that the difference between the result of the exact model and the *meta*-heuristic approaches is less than 5 %, thereby, confirming the quality of solution approaches.

Further, to identify the best solution approach out of four, they were compared in terms of six different metrices. The result from 36 test instances shows that NSGA-II and NSGA-III provide high-quality solutions compared to MOGWO and MOICA. NSGA-III appears better at exploring solution space than NSGA-II. However, NSGA-II is better in terms of the number of Pareto solutions and solution time. Finally, a sensitivity analysis was conducted to ascertain the effect of varying the weights assigned to each criterion on the model outcomes and to draw useful managerial insights.

Even though the tourism industry can ensure significant economic development, research has revealed that the industry has negative environmental and social impacts (Northcote & Macbeth, 2006). Moreover, people are increasingly concerned about the impact of their behavior on the environment even during their entertainment activities (Piya et al., 2022). Therefore, the concept of sustainability can be integrated to generate the so-called green touristic itineraries for the OPTP. Finally, the effect of lockdown periods and traveling disruptions due to exceptional natural and health phenomena on the tourism industry of a city such as Muscat should be studied and assessed (Hosseini et al., 2021).

CRediT authorship contribution statement

Sujan Piya: Conceptualization, Methodology, Writing - original draft, Writing - review & editing. **Chefi Triki:** Writing – original draft, Conceptualization, Writing – review & editing. **Abdulwahab Al Maimani:** Methodology, Writing - original draft. **Mahdi Mokhtarzadeh:** Writing – original draft, Formal analysis.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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